KASSTA Webinar

Korean-American AeroSpace Science and Technology Association

Friday, April 24th, 2026 (virtual)

4:00 PM (PT), 7:00 PM (ET)

Register for the Webinar

Speakers

Woosang Park
Postdoctoral researcher, Department of Aerospace Engineering, Texas A&M University

Dynamic Nonlinearity and Model Fidelity Effects on Uncorrelated Track Processing

Identifying and cataloging space objects is increasingly difficult as thousands of satellites are launched each year and debris accumulates from launches and on-orbit breakups. This growth produces a high volume of daily observations, known as tracks. Tracks not matched to entries in the Space Object Catalog are classified as uncorrelated tracks (UCTs), making automated track association essential for managing them. Effective UCT processing is critical for maintaining catalog accuracy and supporting conjunction assessment and collision avoidance.

One such method is covariance-based track association (CBTA). This work examines two key questions affecting CBTA performance: which orbital element representations minimize dynamical nonlinearity and delay degradation of the association metric, and how dynamic model fidelity influences performance. The results provide insight into the roles of nonlinearity and model fidelity in CBTA reliability.

Hyomin Kim
Professor, Department of Physics, Center for Solar-Terrestrial Research, Institute for Space Weather Sciences, New Jersey Institute of Technology

Space Weather: How Sun–Earth Interactions Affect Human Technology

The interaction between the continuous stream of particles and magnetic fields from the Sun (the solar wind) and Earth's magnetic field creates a unique geomagnetic structure known as the Earth's magnetosphere. Through this system, solar wind plasma transports mass, momentum, and energy into the coupled magnetosphere–ionosphere system, driving geomagnetic activity in the near-Earth environment. The coupling between the solar wind and geospace leads to complex dynamics, including current systems, plasma convection, particle acceleration and precipitation, and magnetic reconnection. Variations in the Sun's energy output and magnetic field such as solar flares and coronal mass ejections (CMEs) disturb the interplanetary environment. These disturbances can interact with Earth's magnetosphere and its upper and middle atmosphere, causing significant disruptions to technology, satellites, power grids, and radio communications, while also producing auroras. This presentation highlights recent efforts within the space science community to better understand these processes.

It is planned to give gift cards to selected participants. To be qualified, join KASSTA and choose KASSTA as your APS if you are a KSEA member.

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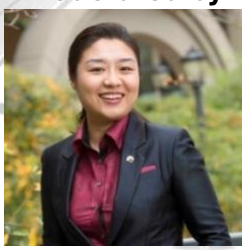

Soyeon Yi
KASSTA Public Relations Director
Former Astronaut